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OPNAVINST 9420.2B
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OPNAV INSTRUCTION 9420.2B

From: Chief of Naval Operations

Subj: NAVY ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEM AND
ELECTRONIC CHART SYSTEM POLICY AND STANDARDS

Ref: (a) CNO ltr Ser N00/8U5000076 of 17 March 1998 (NOTAL)
(b) CNO Executive Board (CEB) Decision Memorandum of 27 November 2000,
Navigator of the Navy CEB (NOTAL)
(c) International Maritime Organization Resolution MSC.232(82), Revised Performance
Standards for ECDIS
(d) NATO STANAG 4564A
(e) COMUSFLTFORCOM/COMPACFLTINST 3530.1C
(f) SECNAVINST 5000.2G
(g) NAVSEAINST 9420.4B
(h) COMNAVSURFPAC/COMNAVAIRPAC/COMNAVAIRLANT/
COMNAVSURFLANT INSTRUCTION 3530.4G
(i) CORIVFORINST 3530.1
(j) DASN Ships Memorandum to Program Executive Office, Integrated Warfare
Systems (PEO IWS), Electronic Chart Display Information Systems-Navy
(ECDIS-N) as an Acquisition Program Under PEO IWS, 24 January 2012 (NOTAL)

Encl: (1) Navy Commissioned Vessel Certification for Unrestricted Operations Two Phase
Process
(2) Navy Non-Commissioned Craft Electronic Chart Systems (ECS) Unrestricted
Operations Process

1. Purpose

a. To provide Navy policy and guidance governing the implementation and operational use of the Navy Electronic Chart Display and Information System (ECDIS), commercial ECDIS and Electronic Chart Systems (ECS).

b. To direct and guide the Navy's transition from paper nautical charts to an electronic charting environment for primary navigation and piloting.

c. To establish the minimum performance standards and the certification process for Navy ECDIS and ECS.

d. To assign responsibilities to specific Navy organizations for requirements development, acquisition, installation, testing, certification, training and sustainment of Navy ECDIS and ECS.

e. This revision establishes policy, updates responsibilities, more clearly articulates core functions and delineates relationships to certain Navy organizations related to the implementation of Navy ECDIS, commercial ECDIS and ECS. This instruction is a major revision and should be reviewed in its entirety.

2. Cancellation. OPNAVINST 9420.2A.

3. Scope and Applicability. This policy applies to all U.S. Navy commissioned vessels, non-commissioned craft (e.g., United States naval ships, amphibious warfare type craft and expeditionary craft) and organizations that support the development, acquisition, testing, installation, training, certification, operation and sustainment of Navy ECDIS, commercial ECDIS and ECS.

4. Background

a. Per references (a) and (b), the initial Navy implementation of ECDIS (i.e., ECDIS-N) has been implemented on the majority of U.S. Navy commissioned ships and submarines. A revision to the current Navy ECDIS policy guidance is warranted to capitalize on the lessons learned from ECDIS-N implementation, exploit advancements in ECDIS technology, further clarify commercial ECDIS and ECS requirements for non-commissioned platforms, standardize ECDIS acquisition across the fleet and maintain interoperability with allied navies.

b. The National Geospatial-Intelligence Agency (NGA) is transitioning away from Digital Navigation Charts (DNC) that are unique to the U. S. Navy and adopting international Electronic Navigation Chart (ENC) standards. Additionally, NGA and other foreign Hydrographic Offices are in-progress or have planned cancellations of their Standard Nautical Chart (i.e., paper chart) portfolio which requires the Navy to fully transition to electronic navigation.

c. The Navy ECDIS Program of Record (POR) was established by Program Executive Office, Integrated Warfare Systems (PEO IWS) to reduce the proliferation of ECDIS hardware and software versions and resultant training and sustainment challenges and to standardize implementation of the next generation of Navy ECDIS across the commissioned fleet. The original implementation of ECDIS (ECDIS-N) will be replaced by the POR Navy ECDIS application software.

5. Policy

a. Commissioned Navy Ships and Submarines

(1) All existing and planned new construction Navy ships and submarines, except those excluded because of planned decommissioning, must use an approved Navy ECDIS POR as the primary navigation plot or must receive a waiver from Office of the Director, Oceanographer and Navigator of the Navy (OPNAV N2N6E) for each affected platform or platform class. Waiver requests must be submitted to the Navy ECDIS Integrated Product Team (IPT) with reasons for non-compliance and an executable plan to become compliant.

(2) All legacy ECDIS configuration approvals and certifications completed under the provisions of this instruction must remain in effect until the legacy ECDIS-N configuration is transitioned to the Navy ECDIS POR for each platform. Responsibilities outlined in subparagraphs 7b through 7e, and 7j which pertain to Navy ECDIS also pertain to ECDIS-N when Navy ECDIS has not been approved for the platform type.

(3) Minimum performance standards for Navy ECDIS:

(a) ECDIS performance standards identified in reference (c).

(b) Warship ECDIS software performance standards identified in annex A of reference (d).

(c) Process and display ENC and submarine electronic navigational charts (SMENC) in the International Hydrographic Organization (IHO) standards for the S-57 and future S-100 data format (when available) series.

(d) Process and display vector product format (VPF) electronic charts and data from the NGA (e.g., DNC, tactical ocean data (TOD), VPF database update and the geospatial symbols for digital displays). These performance standards will be canceled once NGA officially completes production and dissemination of the global portfolio of ENC required by the Services and CCMDs and all commissioned vessels have an ENC-compliant version of ECDIS.

(e) Process, display, transfer and interrogate additional military layers (AML) containing specific military information (e.g., maritime boundaries, Q-routes, Operational Area (OPAREA)).

(f) Accept and display surface search navigation radar target and video overlay, visual and other navigation fix information (e.g., Automatic Identification System).

(g) An independent backup ECDIS matching the platform's ECDIS version, as approved by OPNAV N2N6E, must be provided to ensure safe navigation in the event of primary ECDIS failure.

(h) Display, process and store classified data on all consoles commensurate with the mission security requirements of the host vessel.

b. Military Sealift Command (MSC)

(1) MSC government-owned ships will utilize ECDIS compliant with International Maritime Organization (IMO) performance standards established for civil shipping per reference (c).

(2) MSC vessels of appropriate tonnage may utilize ECS under applicable Federal Regulations.

c. Naval expeditionary craft

(1) All non-commissioned amphibious warfare type craft and Navy expeditionary craft identified in reference (e) must be able to safely navigate using an ECS that meets the performance standards in subparagraph 5c(2) and can process and display electronic charts provided by NGA without reliance on paper charts as a backup. Applicable OPNAV Resource Sponsor and PEO Program Office will provide concurrence of craft type in reference (e).

(2) Minimum performance standards for Navy ECS:

(a) Process and display ENC in the IHO standards for the S-57 and S-100 data format (when available) series.

(b) Process, display, transfer and interrogate AML containing specific military information (e.g., maritime boundaries, Q-routes, OPAREAs).

(c) Craft that are employed in any mission in which direct action with an adversary is reasonably expected are required to employ both a primary and an independent secondary ECS to ensure safe navigation.

(d) Craft that are employed in any mission in which direct action with an adversary is unlikely are only required to employ a primary ECS.

6. Action

a. OPNAV N2N6E establish and chair the Navy ECDIS IPT, comprised of representatives from the OPNAV platform resource sponsor; Operational Test and Evaluation Force (OPTEVFOR); Commander, Naval Sea Systems Command (COMNAVSEASYS COM) PEO IWS Program Office; COMNAVSEASYS COM Navigation Systems-Ships Technical Warrant Holder; Commander, U.S. Fleet Forces (COMUSFLTFORCOM); and platform type commander (TYCOM). The purpose of the IPT is to:

(1) Review and approve the test strategy for candidate Navy ECDIS configuration(s).

(2) Review Navy ECDIS configuration test results and make recommendation to OPNAV N2N6E for Navy ECDIS configuration approval.

(3) Review Navy ECDIS waiver requests and make recommendations to OPNAV N2N6E.

b. If Navy ECDIS is being delivered as part of an acquisition program requiring testing per reference (f), the acquisition program manager is responsible for incorporating the Navy ECDIS certification requirements into the program's delivery schedule and test planning processes.

c. To certify a commissioned Navy vessel for unrestricted operations using a Navy ECDIS as the primary navigation plot, candidate configurations must successfully complete the two phase process described in enclosure (1). To approve a non-commissioned Navy craft for unrestricted operations using an ECS, candidate configurations must successfully complete the process described in enclosure (2).

d. Changes to approved Navy ECDIS baseline configurations are required to be re-approved if determined by the COMNAVSEASYS COM (SEA 05W) Navigation Systems-Ships Technical Warrant Holder, PEO IWS Program Office or OPNAV N2N6E that these changes significantly impact the previously approved configuration. ECDIS baseline configuration changes consist of any major upgrade or major modification of Navy ECDIS equipment and interfaces with the Navy ECDIS. Equipment interfaces considered part of the ECDIS baseline configuration are, but not limited to, positioning, navigation and timing distribution systems; global positioning satellite receivers; inertial navigation systems; gyrocompass; speed sensor; fathometer; connected radar; ECDIS software; and ECDIS hardware. Proposed changes determined to significantly impact the previously approved configuration will be forwarded to the ECDIS IPT for consideration prior to installation. The ECDIS IPT will review updated baseline documentation to support approval of an ECDIS baseline configuration change.

7. Responsibilities. Specific commands and agencies must execute their duties as described in subparagraphs 7a through 7j.

a. OPNAV N2N6E

(1) Collect, review, submit and update Navy ECDIS requirements for validation per reference (f). Requirements must reflect the minimum performance standards of this instruction in conjunction with fleet identified capability gaps.

(2) Resource the development, integration and testing of the Navy ECDIS capability on surface platforms per reference (f).

(3) Establish electronic navigation policy and minimum performance standards for Navy ECDIS.

(4) Approve or disapprove operational use of Navy ECDIS configuration(s) based on review of test results and recommendations provided by the Navy ECDIS IPT. Also, serve as the waiver approval authority for candidate Navy ECDIS configuration(s) not fully compliant with this instruction.

(5) Serve as the primary U.S. Navy liaison to NGA for safety of navigation products and services required to support Navy ECDIS.

(6) Coordinate joint, interagency and international data standards to ensure Navy ECDIS compatibility and interoperability.

(7) Convene and chair the Navy ECDIS IPT.

(8) Maintain the currency of the Navy ECDIS policy, performance standards and certification processes outlined in this instruction.

b. Designated OPNAV Platform Resource Sponsor

(1) Resource the procurement, certification, training, sustainment and modernization of Navy ECDIS POR for commissioned vessels under their cognizance. Modernization includes the requirement to periodically update Navy ECDIS hardware, software and interfaces to new navigation sensors to process and display the latest IHO format standards.

(2) Resource the research, development and testing per approved Navy ECDIS requirements documentation for submarine platforms.

(3) Provide representative(s) to the Navy ECDIS IPT.

(4) Ensure new construction commissioned ship programs acquire and install OPNAV-approved Navy ECDIS software.

(5) Ensure pre-existing and new construction non-commissioned craft programs identified in reference (e) acquire and install or provide ECS that meet the minimum performance standards in subparagraph 5c(2).

c. COMUSFLTFORCOM and Commander, Pacific Fleet (COMPACFLT) (or their designated TYCOM representatives)

(1) Authorize the use of an OPNAV-approved Navy ECDIS as the primary navigation plot on a commissioned ship or submarine based upon:

(a) Installation of an approved Navy ECDIS configuration.

(g). (b) Successful completion of a navigation certification (NAVCERT) per reference

(c) Successful completion of a Navy ECDIS crew certification and navigation assessment that verifies personnel operating the Navy ECDIS can safely navigate the vessel per current TYCOM navigation policy, doctrine and standard operating procedures.

(2) Determine if the Navy ECDIS POR correctly and adequately supports the fleet navigation mission. If capability gaps are identified, consolidate new requirements and forward to OPNAV N2N6E for inclusion into the Navy ECDIS requirements documents.

(3) Authorize the use of an OPNAV-approved ECS as the primary navigation plot on non-commissioned craft identified in reference (e) based upon:

(a) Installation of an approved ECS configuration.

(b) Successful completion of a joint materiel readiness inspection per references (h) and (i) conducted on craft type identified in reference (e), as concurred to by applicable OPNAV Resource Sponsor and PEO Program Office, to ensure ECS operability by PEO Program Office and TYCOM.

(c) Successful completion of an ECS TYCOM conducted crew certification and navigation assessment that verifies personnel operating the ECS can safely navigate the non-commissioned craft per current TYCOM navigation policy, doctrine and standard operating procedures.

d. PEO IWS Program Office

(1) Ensure the minimum Navy ECDIS performance standards of this instruction are incorporated into the approved Navy ECDIS POR acquisition documentation and addressed in the operational test (OT) required by reference (f).

(2) Maintain Navy ECDIS POR per the guidance outlined in reference (j) for development, acquisition and sustainment of the Navy ECDIS program.

(3) Determine the minimum hardware requirements for successful operation of approved Navy ECDIS software version(s).

(4) Coordinate with designated OPNAV platform resource sponsor and the fleet to minimize the number of Navy ECDIS baseline software versions fielded to maximize user interoperability, reduce training requirements and facilitate configuration control.

(5) Coordinate with COMNAVSEASYSCOM (SEA 05H) Integrated Warfare Systems Engineering to perform a NAVCERT on approved Navy ECDIS configurations per reference (g).

(6) Provide a representative for, and responsible to, the Navy ECDIS IPT to formulate and distribute the recommended test strategy for candidate Navy ECDIS configurations requesting approval.

(7) Provide to the Navy ECDIS IPT a software element certification letter for hardware and software configurations based on the PEO IWS-approved element certification plan.

(8) Provide to OPNAV N2N6E waiver approval or disapproval recommendation for candidate Navy ECDIS configurations not fully compliant with this instruction.

e. Participating Acquisition Resource Managers Delivering Navy ECDIS

(1) Coordinate with PEO IWS to ensure timely delivery, integration and testing of OPNAV-approved Navy ECDIS software as well as to incorporate platform-specific requirements.

(2) Incorporate the Navy ECDIS certification requirements into the ship acquisition program's delivery schedule and test planning processes as required by this instruction and reference (g).

(3) Coordinate with COMNAVSEASYSCOM (SEA 05H) Integrated Warfare Systems Engineering to ensure that a NAVCERT is conducted on candidate Navy ECDIS configurations per reference (g).

(4) Provide representative(s) to the Navy ECDIS IPT.

f. PEO Program Office

(1) Ensure the minimum performance standards of this instruction are met.

(2) Coordinate with resource sponsors and the fleet to minimize the number of software versions fielded to maximize user interoperability, reduce training requirements and facilitate configuration control.

(3) Coordinate with TYCOM to perform a joint materiel readiness inspection on ECS to ensure operability.

g. NGA

(1) Provide VPF products (i.e., DNC, TOD, VPF data update and geospatial symbols for digital displays) until the approved sunset date for each product line.

(2) Produce a global portfolio of ENC, SMENC and AML based on the IHO S-57 specification to replace the existing DNC and TOD portfolios. Coordinate with Navy on the transition from ENC based on S-57 to ENC based on the S-100 series of data specifications when published by IHO.

(3) Develop and maintain S57 and emerging S100 series testbed software to validate S57, and when available new and emerging S100 series data products, prior to release to Navy.

h. OPTEVFOR

(1) If the Navy ECDIS capability is being delivered as part of an acquisition program requiring testing per reference (f), ensure requirements of this instruction are properly taken into account during test planning, execution and reporting.

(2) Provide a testing recommendation for candidate Navy ECDIS configuration(s) to the Navy ECDIS IPT. In cases where OT is warranted, determine the scope of OT necessary to sufficiently test for cyber survivability, operational effectiveness and suitability of the Navy ECDIS being installed on the host navigation configuration.

(3) When properly resourced and scheduled, evaluate candidate Navy ECDIS configuration(s). Except for testing that supports an interim approval decision, a recommendation to OPNAV N2N6E regarding configuration approval as a Navy ECDIS must be included in the final test results. Results of testing conducted in support of an interim approval will include a risk assessment regarding the use of the unapproved configuration until the OT required to support an approval recommendation can be completed.

(4) Provide a representative to serve on the Navy ECDIS IPT.

i. Commander, MSC. Establish an acquisition and certification program to transition all government-owned ships supporting Navy missions from paper nautical charts to an IMO compliant ECDIS as the primary navigation plot.

j. COMNAVSEASYS COM (SEA 05). Oversee the overall Navy ECDIS navigation system certification process. The Navigation Systems – Ships Technical Warrant Holder will provide a representative to the Navy ECDIS IPT.

8. Records Management

a. Records created as a result of this instruction, regardless of format or media, must be maintained and dispositioned per the records disposition schedules located on the Department of

the Navy Assistant for Administration, Directives and Records Management Division portal page at <https://portal.secnav.navy.mil/orgs/DUSNM/DONAA/DRM/Records-and-Information-Management/Approved%20Record%20Schedules/Forms/AllItems.aspx>.

b. For questions concerning the management of records related to this instruction or the records disposition schedules, please contact the local records manager or the OPNAV Records Management Program (DNS-16).

9. Review and Effective Date. Per OPNAVINST 5215.17A, OPNAV N2N6E will review this instruction annually around the anniversary of its issuance date to ensure applicability, currency and consistency with Federal, Department of Defense, Secretary of the Navy and Navy policy and statutory authority using OPNAV 5215/40 Review of Instruction. This instruction will be in effect for 10 years, unless revised or cancelled in the interim, and will be reissued by the 10-year anniversary date if it is still required, unless it meets one of the exceptions in OPNAVINST 5215.17A, paragraph 9. Otherwise, if the instruction is no longer required, it will be processed for cancellation as soon as the need for cancellation is known following the guidance in OPNAV Manual 5215.1 of May 2016.



K. O. THOMAS
Deputy Chief of Naval Operations
Information Warfare

Releasability and distribution:

This instruction is available for release and is available electronically only via Department of the Navy Issuances website, <https://www.secnav.navy.mil/doni/default.aspx>.

NAVY COMMISSIONED VESSEL CERIFICATION FOR UNRESTRICTED OPERATIONS
TWO PHASE PROCESS

1. Phase I (blocks 1 through 4). Describes the process required to test, evaluate and approve the Navy ECDIS configuration for the intended host platform(s).

a. Block 1. At the beginning of phase I (Navy ECDIS configuration approval), candidate Navy ECDIS configuration(s) will be identified to OPNAV N2N6E by the OPNAV platform resource sponsor. Configuration(s) will be identified by ship class or submarine class; installed navigation equipment (controls and sensors, navigation data distribution and electronic charting processing and display); power distribution; electronic charting software; and ship control application planned for installation.

b. Block 2. OPNAV N2N6E must convene a Navy ECDIS IPT for each candidate Navy ECDIS configuration. At least 1 month prior to convening the IPT, members will be provided the necessary information from the requesting organization to formulate a recommended test strategy. Information must include, at a minimum, a description of the unapproved configuration; relevant test history; proposed test strategy; and concept of operations describing the Navy ECDIS hardware and software configuration and system architecture. Test strategy must ensure configuration is adequate to support robustness, survivability and the environmental qualifications commensurate with platform requirements to accomplish ship's mission(s). Additionally, test strategy should consider the manpower, personnel, training, maintenance and logistic infrastructure available to support this plan. Each member of the Navy ECDIS IPT must review the documentation and recommendations for the Navy ECDIS candidate configuration to determine if additional testing (NAVCERT assessment, OPTEVFOR, etc.) is required; or provide concurrence on presented test approach to obtain configuration approval; or provide justification for deviation from briefed proposals.

c. Block 3. Each member of the Navy ECDIS IPT must review the documentation and provide a recommendation for configuration approval. Following consensus by the ECDIS IPT, recommendation for or against configuration approval, along with any requests for deviations, must be forwarded to OPNAV N2N6E for consideration. If concurrence on the implementation details of the selected option cannot be achieved, OPNAV N2N6E will make a decision on the issues for which the IPT members cannot reach agreement. In the event an OT is part of the recommended test strategy, OPTEVFOR will specify the testing required to assess the cyber survivability, operational effectiveness and operational suitability of the candidate Navy ECDIS configuration.

d. Block 4. OPNAV N2N6E must review the recommendation(s) and issue a message approving the Navy ECDIS configuration or direct corrective action and additional testing, as appropriate.

2. Phase II (blocks 5 through 8). Lists the systems command and fleet processes required to properly install, test, certify and ultimately authorize the ship to navigate solely using the Navy ECDIS configuration.

a. Block 5. At the beginning of phase II (shipboard ECDIS and crew certification), each approved Navy ECDIS configuration installed aboard a ship or submarine must complete a system operational verification test per COMNAVSEASYSCOM technical specifications. System operational verification test ensures system hardware and software work properly and that interface data is received into the system in the proper format.

b. Block 6. To ensure system hardware, software and interfaces work properly in the new configuration, a NAVCERT must be conducted per reference (g).

c. Block 7. After installation of an approved configuration, and completion of the NAVCERT, individual ships and submarines must successfully complete a Navy ECDIS crew certification, per guidance issued by the respective fleet commander and TYCOM.

d. Block 8. Upon successful completion of the Navy ECDIS crew certification, the TYCOM will authorize the ship or submarine for unrestricted navigation operations using Navy ECDIS as the primary official navigation plot.

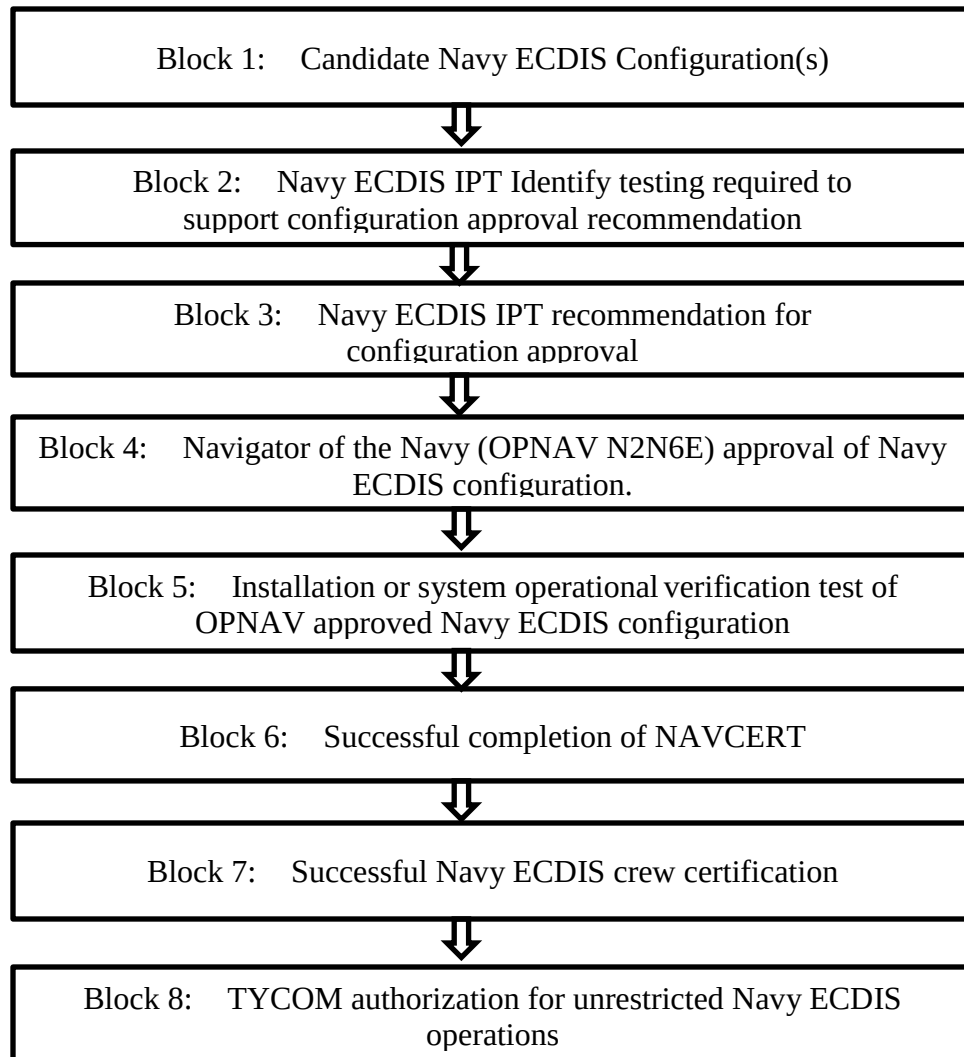


Figure 1

NAVY NON-COMMISSIONED CRAFT ELECTRONIC CHART SYSTEMS (ECS)
UNRESTRICTED OPERATIONS PROCESS

1. Block 1. The ECS installed on non-commissioned craft identified in reference (e) will be selected by the PEO Program Office responsible for procuring the ECS on the specific non-commissioned craft. The ECS selected for non-commissioned craft must meet the minimum standards identified in this instruction. The procuring PEO Program office should seek concurrence that the ECS is suitable and effective for the craft type from the TYCOM and applicable OPNAV Resource Sponsor.
2. Block 2. Each approved ECS installed aboard a non-commissioned craft identified in reference (e) must successfully complete a joint materiel readiness inspection per references (h) and (i) conducted by the installing PEO Program Office and the TYCOM to ensure ECS operability.
3. Block 3. After successful installation and PEO Program Office's TYCOM inspection, the individual non-commissioned craft must complete a successful ECS TYCOM conducted crew certification and navigation assessment that verifies personnel operating the ECS can safely navigate the non-commissioned craft per current TYCOM navigation policy, doctrine and standard operating procedures.
4. Block 4. Upon successful completion of the ECS crew certification and navigation assessment, the TYCOM will authorize the non-commissioned craft for unrestricted navigation operations using the installed ECS as the primary official navigation plot.

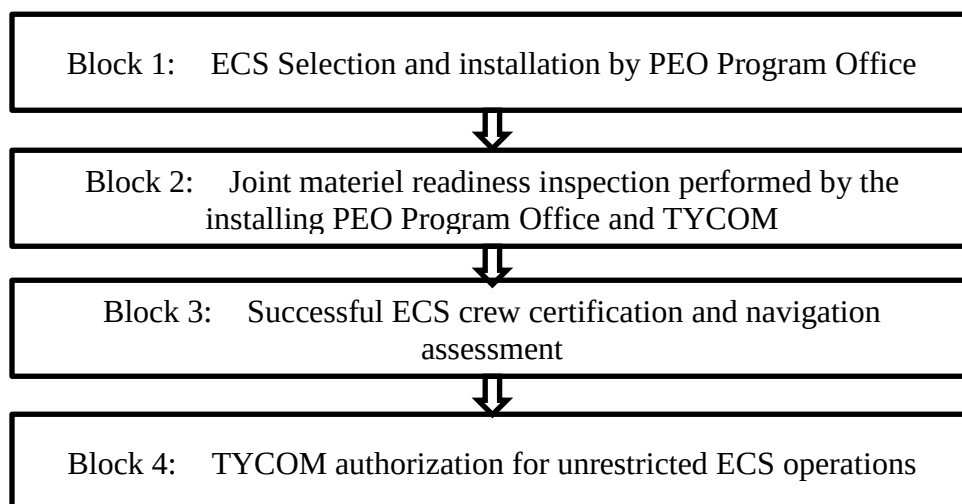


Figure 2